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The use of atomic force microscopy to quantify membrane surface electrical properties

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Abstract

An atomic force microscope has been used to quantify the electrical double layer interactions between a silica particle and a Desal nanofiltration membrane in electrolyte solutions of various ionic strengths. The experimental data has been compared to DLVO predictions incorporating a numerical solution of the non-linear Poisson–Boltzmann equation. At the lowest ionic strength studied (10^{-3} M) there was good agreement between the experimental data and the theoretical calculations. The membrane surface potential calculated from the best-fit membrane charge under these conditions was in good agreement with the zeta potential obtained from streaming potential data. The discrepancy between DLVO calculations and experimental data increased as the ionic strength increased and in particular as the membrane surface roughness became greater in magnitude than the solution Debye length. Under such conditions, colloid–membrane electrical double layer interactions may at present only be quantified by direct experimental measurement. Atomic force microscopy measurement of pull-off (adhesion) forces provides a further direct evaluation of fouling propensity. © 2002 Elsevier Science B.V. All rights reserved.

Keywords: Atomic force microscopy; Desal nanofiltration membrane; Theoretical calculations

1. Introduction

Membrane separation processes are extensively used in industrial and medical applications to separate ions, colloids and biological molecules. For optimum operation the membrane has to possess physical properties giving appropriate interactions with solutes in the process stream. The most important of these properties are pore size distribution, surface morphology, appropriate electrical double layer interactions and minimum adhesion (fouling). Surface interactions may be explained in the first instance within the framework of the DLVO theory [1] by the interplay between two forces: the London-van der Waals force and the electrical double layer force.

The surface electrical properties of membranes have been most commonly evaluated by electrokinetic techniques such as streaming potential measurements [2]. For smooth, homogeneous surfaces this method allows calculation of a surface poten-

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tial (zeta potential) that may be used in subsequent quantification of the interaction between colloidal materials and the surface through the DLVO theory. However, for materials with surface chemical inhomogeneities or significant roughness this method only allows calculation of some average value of an electrical potential at some unspecified shear plane. Many membrane materials have significant surface roughness, so such an approach may not be appropriate for quantitative evaluation of membrane–colloid interactions and hence membrane fouling.

Recently, atomic force microscopy (AFM) [3] has been increasingly used to measure DLVO [4–7] type interactions between a single colloid particle and normally a flat surface as a function of separation distance. Such studies have made use of the colloid probe technique in which a colloidal particle is immobilised at the end of the AFM cantilever. This approach allows direct measurement of colloidal interactions with the advantage of uniquely allowing quantification of local variations in such interactions. AFM has proved a useful tool for membrane technologists as it enables direct measurement of the key physical properties controlling membrane performance [8]. For example, adhesion (pull-off forces) of different types of colloid probes on different membranes have been measured [10–12] directly. Such measurements allow an assessment of the fouling propensity of membranes without the need for process measurements. The colloidal probe technique has also been used to quantify the variations in electrical double-layer interactions and adhesion at different locations on a very rough membrane surface [12]. However, due to the extreme surface roughness it was not possible to calculate surface potential or charge from the experimental data in that case.

The quantitative evaluation of surface electrical properties (charge and potential) from AFM force–distance measurements requires a numerical solution of the non-linear Poisson–Boltzmann equation (PBE). The PBE has normally been solved under the constant charge or constant potential boundary conditions. Previous work incorporating quantitative analysis [13,14] of sphere–smooth surface interactions has reported calculation of the surface charge or surface potential from force–distance data using the Debye length as a fitting parameter. For example, if the electrical properties of the colloid probe are known (usually from electrophoretic measurements) it is possible to quantify the electrical properties of the surface by fitting the experimental force–distance data to theoretical calculations.

The present paper reports experimental force–distance data for the approach of a silica colloid probe to a commercial nanofiltration membrane of intermediate roughness. Such data has been quantitatively analysed by comparison and fitting to numerical solutions of the PBE allowing determination of membrane potential or charge. Similar measurements made between a silica colloid probe and a mica surface have been used as a reference point. This allows an assessment of the use of electrokinetic measurements as a means for quantifying colloidal interactions with rough surfaces. Finally, adhesion of the colloid probe to both surfaces has been quantified through AFM measurement of pull-off forces.

2. Theory

The solution of the asymmetric potential profile for two dissimilar flat plates was obtained using a 1D finite difference predictor corrector scheme for the non-linear PBE,

$$\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = \frac{2n^0 ze}{\varepsilon}\sinh\left(\frac{ze\psi}{kT}\right), \tag{1}$$

where n^0 is the electrolyte number density (m^{-3}), z, the valence, e, the elementary electron charge (C), ε, the permittivity of water ($\mathrm{C\ V^{-1}\ m^{-1}}$) at 25 °C, k, the Boltzmann constant ($\mathrm{J\ K^{-1}}$), x, the separation of the two bodies (m), ψ, the surface potential (V) and T, the absolute temperature (K).

The finite difference mesh was set at 1000 nodes which in conjunction with an interpolation subroutine produces a smooth curve and allows experimental data to be well fitted. The overall calculation process was: (i) Solution of the PBE to give the electrostatic gradient and electrostatic potential ($\partial\psi/\partial x, \psi$) for flat plates with appropriate surface potentials or surfaces charges; (ii) Use

of the osmotic pressure method to obtain the electrical double layer interaction energy (V_R); (iii) Calculation of the London-van der Waals interaction energy (V_A) using an appropriate Hamaker constant. A value of 1.0×10^{-20} J is appropriate for both silica–mica and silica–membrane interactions [15]. The magnitude of this interaction was in all cases small compared to the electrical double layer interaction as only separation distances > 1 nm were considered in calculations. (iii) The Derjaguin approximation was used to calculate the normalised force of interaction between a sphere and a planar surface ($F/R = 2\pi V_T$), where F is the force, R, the radius of the sphere and V_T ($= V_R + V_A$) the total interaction energy.

Two procedures of calculation were used in the study: (a) numerical solutions with the measured zeta potential (or charge, using Eq. (2)) of silica, theoretical Debye length (κ) and appropriate ranges of membrane potential or charge; (b) fitting of the membrane surface potential (ψ) or membrane surface charge (σ) by minimisation of the squared residual deviation between the calculated and experimental values (least squares fitting). The best-fit surface charge/potential was found using a grid search method [16]. Implementation of the computational procedure outlined above was carried out by a FORTRAN 77 code working under a Microsoft NT Operating System on a Pentium II 366.

3. Experimental

The membrane used in this study was a composite Desal DK polyamide membrane (Osmonics) which has a molecular weight cut-off in the range 150–300 for uncharged solutes.

AFM was carried out using an Autoprobe CP-100 (Park Scientific Instruments). In order to study the surface morphology of the membranes they were imaged in double layer mode in 10^{-2} M NaCl solution at pH 6, at a scan rate of 1 Hz and 256×256 pixels resolution. The only preparation of membrane samples was their attachment to a steel disk by double scotch tape. The cantilevers used for imaging were Silicon Ultralevers (Thermomicroscopes) with a specified spring constant of 0.29 N m^{-1} and an integrated high aspect ratio tip of end-diameter ~ 10 nm.

Colloid Probes were made by attaching a silica sphere (2.4–2.75 μm, Polysciences) with loctite glass bond (Loctite Ltd) to a standard V-shaped AFM tipless cantilever. The sphere was immobilised by the use of a micromanipulator (Singer Instruments), which allowed precise location of the sphere at the apex of the cantilever. Silica spheres have been shown to have a maximum peak-to-valley roughness of 3 nm over an area of 0.45 μm [4]. AFM allows measurement of the force between the probe and a sample as a function of the displacement, which is varied using a piezoelectric crystal. A laser beam reflected from the back of the cantilever falls onto a split photodiode that detects small changes in the deflection of the cantilever. To convert the deflection to a force it is necessary to know the spring constant of the cantilever and to define the zero of force. The cantilever spring constant can vary substantially from that specified by the manufacturer. For this reason each cantilever used was calibrated directly [17] and the values of spring constant were found to be in the range of 0.34–0.42 N m^{-1}. The zero of force was chosen when the deflection of the cantilever was independent of the piezo position (when the probe and the membrane surface where far apart). Forces reported are normalised by the radius of the sphere attached to the cantilever. It is also important to define the zero of distance at which the colloid probe and membrane come into contact. To define the zero of distance the deflection of the cantilever from its position at large distances was subtracted from piezo movement values, point by point. The plot of this distance (abscissa) against force (ordinate) becomes vertical at the zero of distance—this is also the onset of the constant compliance region, where the output of the photodiode becomes a linear function of the sample displacement. As there remains some uncertainty in the exact position of the zero distance, theoretical fitting to experimental curves was only carried out for data at separation distances > 1 nm. Force–distance measurements between silica–mica and silica–membrane surfaces were made in NaCl solutions

of concentrations 10^{-3}, 10^{-2} and 10^{-1} M at 25 °C and pH 6. The probe movement was kept constant at 0.1 μm s^{-1}.

The zeta potential of the silica spheres was calculated from experimentally determined electrophoretic mobility (Malvern Zetamaster) [18]. The mean values determined in NaCl solutions at pH 6 were: -77.2 ± 4.1 mV at 10^{-3} M; -69.2 ± 3.3 mV at 10^{-2} M; -42.5 ± 2.6 mV at 10^{-1} M.

4. Results and discussion

4.1. Double-layer electrostatic interactions

Mica was chosen as a reference point because of its relatively smooth hard surface and as it has been shown that mica interactions may be accurately modelled by the DLVO theory [19]. Deconvolution of the cantilever deflection against piezo displacement for the approach of a colloid probe to a surface allows quantification of the electrical double-layer interactions as a function of separation distance [20]. Typical data are shown in Fig. 1 for silica–mica interactions in 10^{-3} M NaCl. The data shown is averaged from four sets of data taken at different points on the mica surface. It was found that force approach curves at this electrolyte concentration were highly reproducible. Fig. 1 clearly shows predominant double-layer repulsive interactions. DLVO calculations including theoretical solutions of the PBE for the constant charge and constant potential boundary conditions have been included in Fig. 1. The experimental force curve lies within the limits of the theoretical values, which is expected for a flat mica surface. The experimental profiles for silica–mica at 10^{-2} M were also well modelled with the theory (data not shown). The best fit surface potential (ψ) and surface charge (σ) for mica were $\psi = -82$ mV and $\sigma = -0.0040$ C m^{-2} in 10^{-3} M NaCl and $\psi = -42$ mV and $\sigma = -0.0048$ C m^{-2} in 10^{-2} M NaCl. The calculated potential is in agreement with previous findings [21–26]: reporting electrokinetic potentials for mica in 10^{-3}–10^{-2} M 1:1 electrolyte were in the range -85 to -40 mV.

Surface roughness considerations become important where the range of the surface force is of the same order as the surface roughness. Fig. 2(a) gives a line profile for the mica surface, showing a variation ~ 0.4 nm. The root-mean-square (rms) roughness over an area of 3×3 μm was 0.7 nm. These values are significantly less than the Debye length for 10^{-2} and 10^{-3} M solutions, Table 1.

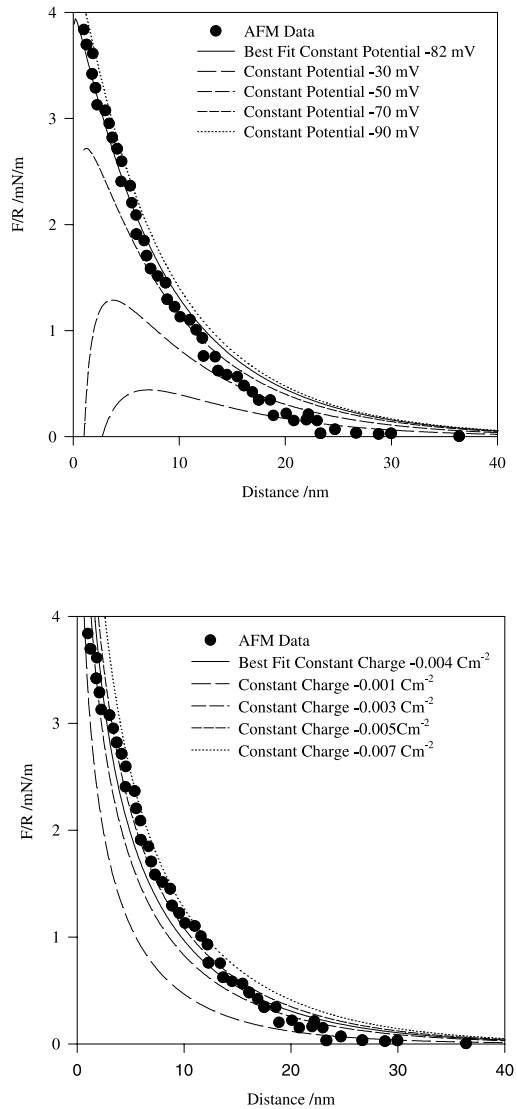


Fig. 1. Silica–mica approach data in 10^{-3} M NaCl with DLVO calculations for constant potential and constant charge boundary conditions.

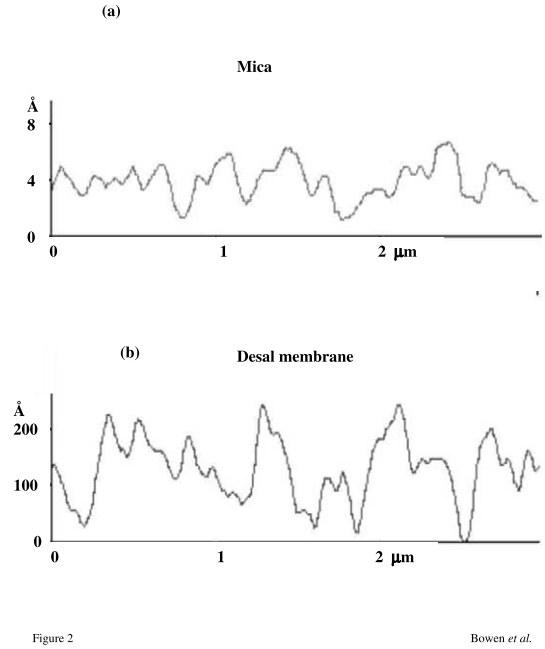


Fig. 2. Typical line profiles of mica and Desal membrane.

Experimental effective decay lengths are shown in Table 1. There is good agreement between these effective decay lengths and the Debye lengths in 10^{-2} and 10^{-3} M solutions. The agreement is less good in 10^{-1} M solution, probably partly due to the difficulty in defining the exact zero of distance and partly due to the effect of non-DLVO interactions at the shorter distances over which measurements were made at such a high salt concentration.

Fig. 3 shows a typical 3D AFM image of the Desal membrane over an area of 3×3 μm, obtained by scanning with a tip. Dark regions are depressions (valleys) in the membrane surface and lightest areas are asperities on the membrane (peaks). Rms roughness over an area of 3×3 μm was 8.5 nm. Thus, the Desal membrane may be classified as membrane of intermediate roughness [8,10]. The measured rms membrane surface roughness is an order of magnitude higher than the rms roughness of mica. The typical line profile for the Desal membrane shown in Fig. 2(b) gives a variation of ~ 20 nm. Comparison of these roughness values to the Debye length shows that the effect of roughness on interactions is in the first instance expected to be apparent for solutions of all of the ionic strengths studied.

Table 1
Debye length and effective decay lengths for approach curves between silica–mica and silica–Desal membrane

NaCl (M)	Debye length (nm)	Effective decay lengths (nm)	
		Mica	Desal membrane
10^{-3}	9.6	9.2 ± 0.5	9.8 ± 1.0
10^{-2}	3.1	3.5 ± 0.3	4.5 ± 0.8
10^{-1}	0.96	1.8 ± 0.2	2.2 ± 1.3

Generally, it was observed throughout the course of the experimental force–distance work with the Desal membrane that a decrease in ionic strength from 10^{-1} to 10^{-3} M resulted in an increase in the reproducibility of the experimental data. Figs. 4–6 shows typical AFM force–distance data (each the average of 10 curves) acquired at different ionic strengths. The measured forces at all electrolyte concentrations were repulsive and their range decreased with increasing ionic strength. For each set of data, DLVO curves have been plotted together with the experimental data. Comparison of Figs. 4–6 shows that the AFM data are closest to the theoretical calculations at 10^{-3} M. With increasing ionic strength the experimental data progressively deviate from the theory. Experimental decay lengths are given in Table 1. There is reasonable agreement with the Debye length in 10^{-3} M solution, but substantial discrepancies with the Debye length in 10^{-2} and 10^{-1} M solutions. Hence, it is only at the lower ionic strength that the interaction of colloid probe and membrane may be reasonably modelled on the basis of a DLVO interaction with a planar surface. We have previously reported the decay lengths for a colloid probe approaching peaks and valleys of membranes with very rough surfaces [9]. However, in the present work the physical size of the silica probe compared to the periodicity of surface roughness prevents it from approaching the membrane surface at a valley bottom.

The experimental force approach curves for mica and the membrane and their theoretical best

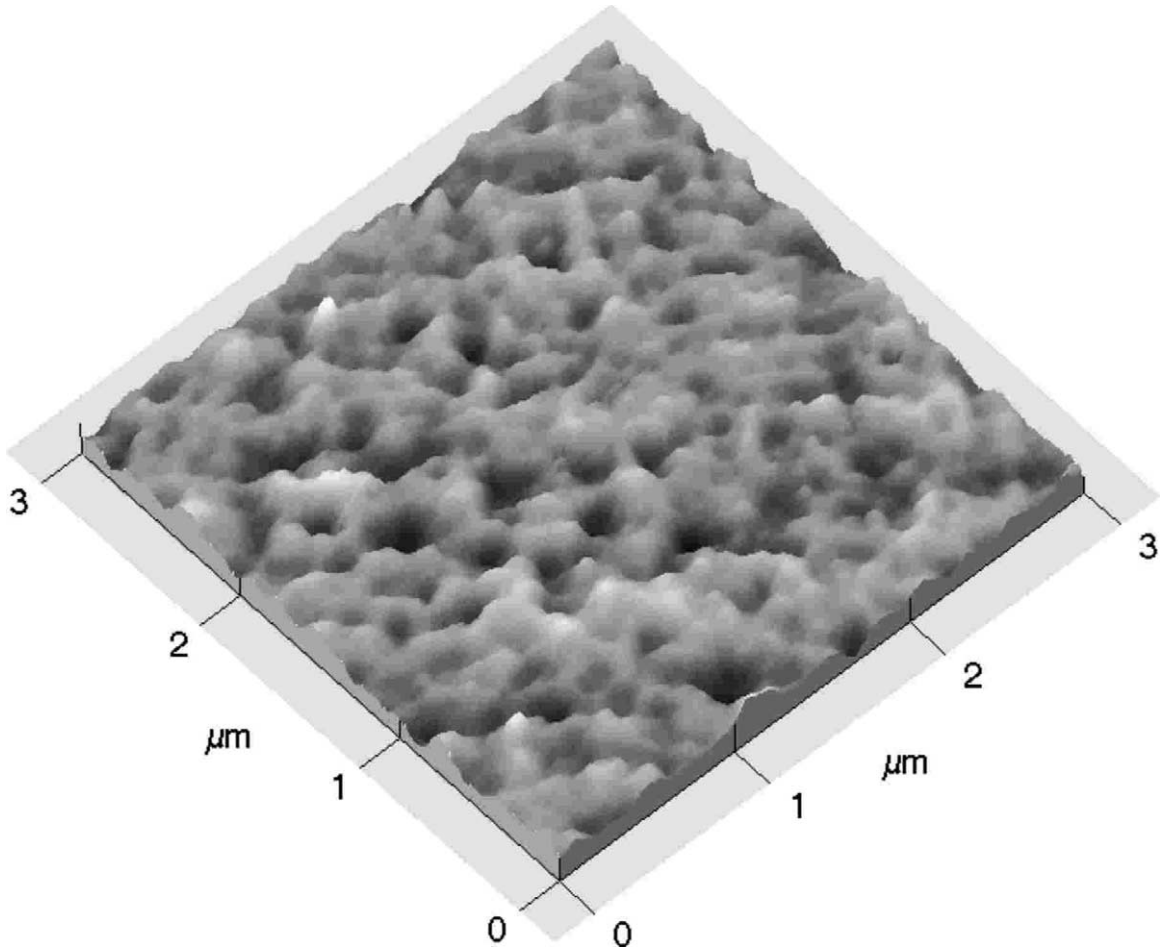


Fig. 3. AFM 3D image of a Desal NF membrane.

fits at 10^{-3} M are compared further in Fig. 7, where the thick and thin lines, respectively represent the constant potential and constant charge limits of the DLVO theory. These cases are the limits of the actual process (charge regulation) which is expected to take place as the interfaces approach. For mica, the experimental data fit well between the constant potential and constant charge curves, Fig. 7(a). For the membrane, at shorter separation distances the data are modelled well by the constant charge approximation, at larger distances the data fit the constant potential approximation better, Fig. 7(b). The observed difference between the best fit constant potential and constant charge curves for the membrane is far greater in the case of the membrane than for mica. It is also worth noting that while most of the data sets for the membrane gave best fit charge densities closely grouped around -0.0010 C m^{-2}, very occasionally substantially different data sets were recorded. An extreme best fit charge density of -0.0051 C m^{-2} was recorded (not included in averaged data). This indicates regions on the membrane with very different surface chemistry or morphology.

The numerical analyses of AFM experimental data for all ionic strengths are summarised in Table 2. The magnitudes of the fitted values of potential increase with decreasing ionic strength, whilst the magnitude of the fitted charge increases

with ionic strength. The latter matches the trend in volume charge density for the membrane, which may be evaluated from mathematical modelling of salt rejection [27,28]. However, care is needed in the use of the values in Table 2 as it was only at the lowest ionic strength that a good match between theory and experiment was obtained.

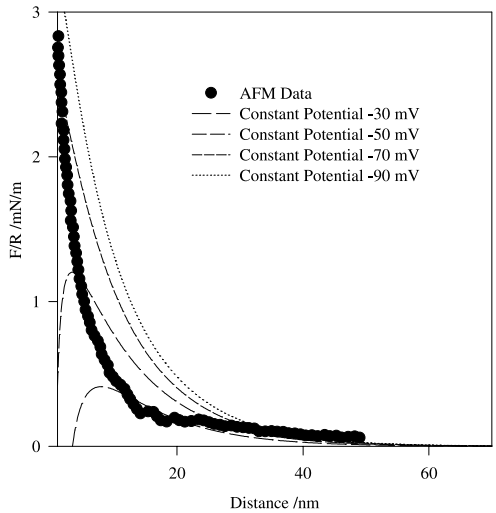


Fig. 4. Silica–Desal membrane approach data in 10^{-3} M NaCl with DLVO calculations for constant potential and constant charge boundary conditions.

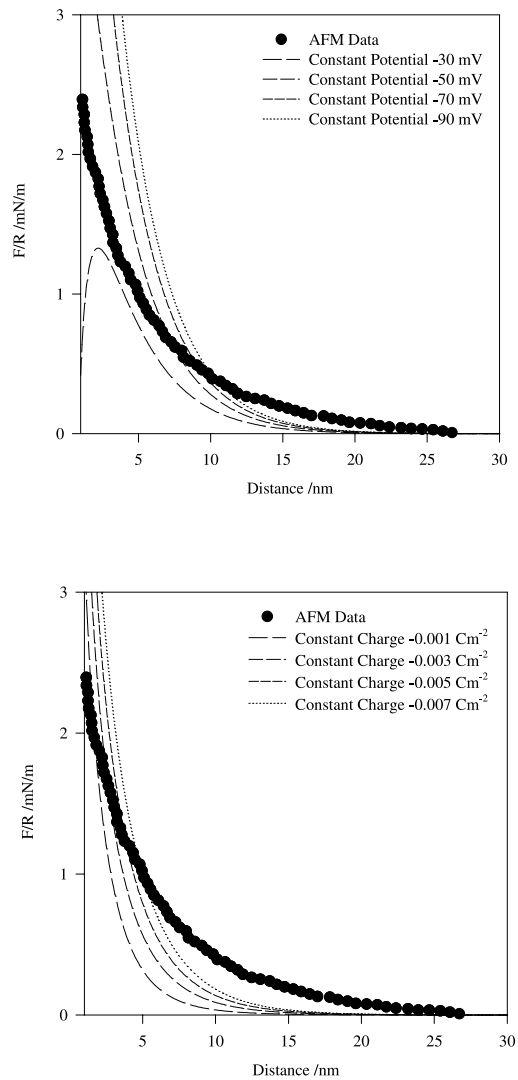


Fig. 5. Silica–Desal membrane approach data in 10^{-2} M NaCl with DLVO calculations for constant potential and constant charge boundary conditions.

For the latter case, the best fit was for the constant charge boundary condition. The best fit charge for this boundary condition may be converted to a potential using,

$$\sigma = \frac{-4n^{o}ze}{\kappa}\sinh\left(\frac{ze\psi}{2kT}\right). \quad (2)$$

This gives a value of −15.8 mV, which may be compared to the value obtained from streaming potential measurements of −15.3 mV [29]. Thus, for the 10^{-3} M solution, not only may the AFM data be reasonably well described by DLVO theory, but such analysis also gives a characterisation of the surface electrical properties in agreement with electrokinetic data. However, such electrokinetic measurements only have clear validity where the surface roughness is small compared to the Debye length. The AFM measurements of interaction show that in other cases it is necessary to make direct experimental measurements if the electrical double layer interactions are to be correctly quantified.

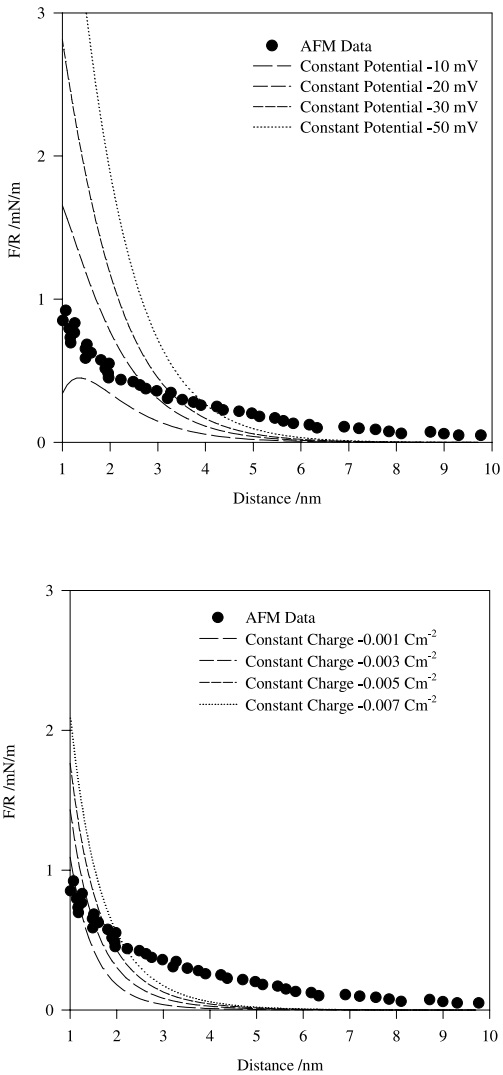


Fig. 6. Silica–Desal membrane approach data in 10^{-1} M NaCl with DLVO calculations for constant potential and constant charge boundary conditions.

4.2. Adhesion (fouling) properties of Desal membrane

From a practical viewpoint, the main reason for quantifying the electrical double layer interactions between a particle and a membrane by AFM, or of electrokinetically characterising a membrane surface, is to quantify the propensity of the surface to foul in use. AFM in conjunction with the colloid probe technique also allows a direct quantification of membrane fouling through the measurement of pull-off (detachment) forces when the probe is retracted from the surface after contact has been made.

Data for such pull-off forces is given in Table 3 for a silica probe interacting with the Desal membrane and also for such a probe interacting with a mica surface. For the Desal membrane, the magnitude of the pull-off force increases systematically with increasing ionic strength, as expected if the electrical double layer repulsive forces are increasingly screened. For mica, the magnitude of the pull-off force increases between 10^{-3} and 10^{-2} M but then shows a small decrease at 10^{-1} M, probably due to the effect of short range forces. The surfaces have pull-off forces of comparable magnitude in 10^{-3} M solution, but the increase with ionic strength is less for the membrane, possibly due to the fact that the effective electrical double layer repulsion is already diminished even at the lowest ionic strength due to the surface roughness. The ratio (SD/Mean F/R) for the Desal membrane is always greater than for mica, a result of the variation in the interactions taking place at different places on the membrane surface. This is clear indication of the effect that roughness distribution can have on the adhesion of foulants to a membrane surface.

These pull-off measurements again point to the advantage of direct measurement if the interactions of particles with rough surfaces, such as membranes are to be quantified. In principle, a

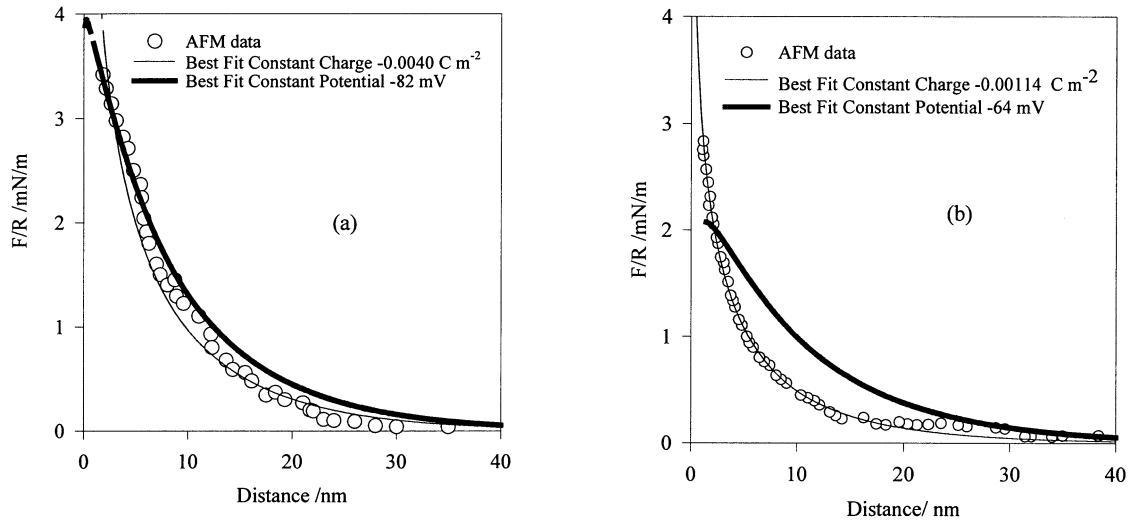


Fig. 7. (a) Silica–mica and (b) silica–Desal membrane interactions in 10^{-3} M NaCl with best fit DLVO calculations for constant charge and constant potential boundary conditions.

theoretical approach which takes into account the real membrane morphology by means of surface element integration is possible [30]. However, for any real surface such an approach is excessively computationally demanding.

5. Conclusions

The present paper has shown how electrical double layer interactions between a colloid probe and a membrane of intermediate roughness may be experimentally quantified by using an atomic force microscope. Experimental force–distance curves have been compared to DLVO calculations incorporating a numerical solution of the non-linear PBE. The best agreement between theory and experiment was obtained at the lowest ionic strength studied (10^{-3} M) and for a constant charge boundary condition. The potential calculated from the best fit charge under these conditions was in good agreement with the zeta potential obtained from streaming potential data.

The discrepancy between DLVO calculations and experimental data increased as the ionic strength increased, and in particular as the membrane surface roughness became greater in magnitude than the solution Debye length. Under such conditions, the interpretation of streaming potential data becomes vague, and calculations of membrane–colloid interactions based on zeta potentials obtained from such electrokinetic data are expected to be in error, in particular as the experimental decay length deviates significantly from the Debye length. For such conditions, AFM uniquely offers the possibility of quantifying directly the interactions. Additionally, direct measurement of pull-off (adhesion) forces allows experimental evaluation of the propensity of the membrane for colloidal fouling without the need for process data.

The AFM data also shows that some areas of the membrane may have substantially different surface chemistry or morphology. This suggests that future research work aimed at quantifying

Table 2
Fitted diffuse layer charges and potentials of Desal membrane

Ionic strength (M)	Best fit charge (C m^{-2})	Best fit potential (mV)
10^{-1}	−0.0040	−16
10^{-2}	−0.0021	−38
10^{-3}	0.00114	−64

Table 3
Adhesion characteristics of the Desal membrane and mica

Surface type	Electrolyte concentration (M)	F/R (mN m^{-1})	Mean F/R (mN m^{-1})[a]	SD/Mean
Desal				
	10^{-3}	Min −0.07 Max −0.8	−0.5 (±0.37)	0.74
	10^{-2}	Min −0.3 Max −1.09	−0.85 (±0.61)	0.72
	10^{-1}	Min −0.4 Max −1.45	−1.12 (±0.78)	0.70
Mica	10^{-3}	Min −0.19 Max −0.83	−0.4 (±0.18)	0.45
	10^{-2}	Min −1.2 Max −3.1	−1.6 (±0.56)	0.35
	10^{-1}	Min −0.95 Max −2.3	−1.3 (±0.52)	0.40

[a] Values in brackets are SD.

membrane-fouling mechanisms should consider the development of mathematical algorithms, which describe local variations in membrane properties.

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